

Neural Networks-Based Wi-Fi/PDR Indoor Navigation Fusion Methods

Da Yu^{ID}, Changgeng Li^{ID}, and Jiaxun Xiao^{ID}

Abstract—Pedestrian dead-reckoning (PDR) technique has high accuracy in short-term positioning and is little affected by environments, but its cumulative error makes long-term positioning infeasible. Received signal strength (RSS) fingerprint positioning is immune to the cumulative error, but its poor radio signal propagation affected by indoor environments leads to a large positioning error. The extended Kalman filter (EKF) is widely applied to improve the two methods, but the approximation of nonlinearity still causes a large localization error. In this study, we propose a new approach based on neural networks for the Wi-Fi/PDR positioning fusion. The nonlinear mapping capability of back-propagation (BP) neural network is used to correct the EKF nonlinear approximation errors. Unlike the commonly used PDR algorithm, a series of pedestrian historical motion states are applied to train the long short-term memory (LSTM) and then make predictions. This method can significantly reduce cumulative errors in PDR. On-site experiments demonstrate that the proposed method achieves an average localization error of 1.18 m with 73% of the errors under 2 m, which outperforms EKF and unscented Kalman filter (UKF) by approximately 45% under the same test environment. Moreover, another fusion method using LSTM localization results as a priori information is described in this article. Different from the fusion at the resulting level of the previous method, this second method achieves the fusion at the algorithm level by limiting the Wi-Fi reference point (RP) search range through the LSTM localization results to reduce training and positioning time.

Index Terms—Indoor fusion localization, long short-term memory (LSTM), pedestrian dead-reckoning (PDR), Wi-Fi positioning.

I. INTRODUCTION

THE development of smartphones and mobile networks makes people's lives more convenient. Location-based service (LBS) [1] has a wide range of application scenarios. Particularly, the indoor positioning system (IPS) is key in a variety of indoor LBS applications, such as emergency rescue [2], smart home [3], smart city [4], and so on.

Due to the obstruction of concrete and other obstacles, the global positioning system (GPS) [5] and other satellite technologies lack precision or fail entirely in indoor environments. Many studies have shown the advantages of radio frequency

(RF) signals, such as infrared technology [6], wireless local area networks (WLANs) [7], Bluetooth [8], RF identification (RFID) [9], ZigBee [10], and ultrawideband (UWB) [11] in indoor positioning. With the widespread deployment of Wi-Fi access points (APs), technologies based on the existing Wi-Fi infrastructures for positioning are receiving great attention. Wi-Fi positioning techniques can be divided into geometric feature-based methods (e.g., time of arrival (TOA) [12], angle of arrival (AOA) [13]) and fingerprint-based methods (e.g., received signal strength (RSS) [14]). Fingerprint-based methods using RSS as a signal source have gained a large amount of attention in recent years because they can provide locations without the knowledge of the location of the APs nor additional hardware deployment. However, there are still challenges of using RSS to achieve high-accuracy positioning.

- 1) Fingerprint database needs to be regularly updated; otherwise, the change of APs will lead to large positioning errors.
- 2) Changes in the indoor environment, the impact of obstructions, and the movement of people can cause fluctuations in RSS, and the orientation of the terminal held by users can also have an impact on RSS [15].
- 3) Additionally, different devices fitted with different Wi-Fi chips and hardware variance can significantly limit the positioning accuracy [16]. These difficulties pose challenges to design high-accuracy Wi-Fi positioning algorithms.

With the advance in micro-electromechanical systems (MEMS) technology, it is possible to integrate inertial sensors (i.e., accelerometers and gyroscopes) and magnetometers in smartphones. Inertial sensors can measure the object's real-time motion, such as acceleration and angular velocity, to obtain motion distance and direction. Pedestrian dead-reckoning (PDR) is the algorithm that calculates the relative positions of pedestrians using inertial sensors [17]. The inertial sensor-based PDR navigation does not depend on additional transmission or reception devices in the environment, which makes positioning more flexible. However, consumer-grade MEMS suffers from significant bias instability and random walk-driven drift [18], and the integration of inertial sensors by PDR makes the drift error. Its positioning accuracy is degraded over time. Thus, PDR can only provide short-term reliable locations.

The advantages and disadvantages coexist when we use only a single technology for positioning. A combination of different positioning technologies is the key to achieve reliable and

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accurate indoor navigation. Since both Wi-Fi and PDR positioning technologies are implemented on consumable portable mobile devices, their integration has been widely studied. Considering the errors of Wi-Fi with small localization bias and large variance and the errors of PDR with large localization bias and small variance, many researchers fused the two techniques with the extended Kalman filter (EKF) [19], [20], [21]. These studies considered the PDR error model as the system model and the Wi-Fi results as the observation model. Because the system model is nonlinear, the EKF performs a linear approximation through a first-order Taylor expansion of the nonlinear model to accomplish the fusion. The linearization of EKF requires the computation of the Jacobian matrix. The neglect of higher order terms in the Taylor expansion can introduce large errors; thus, other researchers designed fusion algorithms with the help of higher precision unscented Kalman filter (UKF) [22], [23]. UKF is similar to particle filter (PF) [24], which applies the unscented transform for linear approximation. However, UKF requires a lot of time to process all sigma points for each sampling, which increases significantly the computational effort. UKF can obtain the desired accuracy only in the case that the posterior distribution is well approximated by the Gaussian distribution. In summary, the Kalman filter is based on the linear Gaussian model, and the accuracy of the prior linearization is limited by the rate of system state evolution. When the pedestrian walking path is complex and changes rapidly (e.g., contains multiple turns), the fusion performance using EKF and UKF is limited.

In this work, we aim to enhance the performance of EKF by using neural networks to design high-accuracy Wi-Fi and PDR fusion methods. The main contributions are summarized as follows.

- 1) We propose a Wi-Fi/PDR fusion positioning method based on neural networks, which jointly considers the advantages of PDR and Wi-Fi localization algorithms and improves EKF fusion ability by using the strong nonlinear mapping capability of back-propagation (BP) neural network to improve localization accuracy.
- 2) Two methods are proposed to reduce the cumulative error of PDR. First, we design an attitude-determination EKF to fuse the inertial sensors and the magnetometer to provide a reliable attitude. Second, by taking advantage of the strong learning ability on time series data of long short-term memory (LSTM), more PDR history data are considered to train an LSTM neural network for localization.
- 3) To reduce the impact of RSS fluctuations on Wi-Fi positioning, we use LSTM positioning results to limit the reference point (RP) matching range to prevent matching to distant RPs, providing more reliable and faster positioning results.

The rest of this article is organized as follows. In Section II, related works are discussed. Section III introduces the methodology, including PDR, Wi-Fi fingerprinting, LSTM, and structures for their integration. Section IV reports the experimental setup and results for the performance evaluation. Finally, Section V concludes this article.

II. RELATED WORKS

A. RSS-Based Wi-Fi Fingerprint Positioning

Wi-Fi fingerprinting algorithms can be divided into deterministic and probabilistic methods. K -nearest neighbor (KNN) [25] and its improvement algorithm weighted KNN (WKNN) [26] are classical deterministic algorithms. They estimate locations by averaging or weighted averaging the top K RPs with the closest distance (Euclidean distance or Manhattan distance [27]). These algorithms are simple, but the accuracy is limited. To deal with more complex and variable indoor environments, much more sophisticated machine learning and deep learning methods (e.g., support vector machine (SVM) [28], [29], artificial neural network (ANN) [30], recurrent neural networks (RNNs) [31]) are used to achieve high-accuracy localization. Neural networks based on BP require more computing resources to optimize parameters. To reduce the training time and computational resource consumption, some scholars used radial basis function (RBF) [32], [33] neural networks to design localization algorithms. The RBF neural network is a local approximation that has a faster learning convergence speed than the BP neural network, but the positioning accuracy is still insufficient. Compared to deterministic methods, probabilistic methods can deal with more complex and variable indoor scenes. In the offline training phase, the probabilistic method infers and saves the statistical distribution of the measured RSS and different researches have made different inferences about the signal distribution. Hours system [34], [35] applied standard Gaussian probability density function (PDF) assumption, and Chen et al. [36] made the double-peak Gaussian distribution (DGD) assumption under some circumstances, but Honkavirta et al. [37] showed that the RSS distribution can be right-skewed, symmetric, or left-skewed. Hence, the PDF assumptions can only represent a part of the real RSS distribution [38]. In our previous work [39], instead of the assumption of Gaussian PDF, the RSS from different flow states was collected to build the fingerprint database. We used the improved probabilistic linear discriminant analysis (PLDA) model to project the RSS into a hidden space containing less noise and more distinctive features. In the online location determination phase, the Bayes rule is often used by probabilistic methods to calculate the probability of which RPs match. Compared with distance, posterior probability can cope with a certain degree of RSS fluctuation and lead to better performance than deterministic methods.

In this article, we use K -PLDA [39] as the Wi-Fi fingerprint localization algorithm and developed it to dynamic scenarios by fusing with PDR.

B. Inertial Sensor-Based Navigation

PDR navigation contains three components: step detection, step-length estimation, and attitude determination. For step detection, Davidson and Takala [40] pointed out that the acceleration variation in the vertical direction is most obvious during walking and can be used for step detection. Yao et al. [41] improved the peak detection method based on dynamic time-bending combined with overzero detection for different motion states of pedestrians, which effectively

improved the detection accuracy. In addition to analyzing vertical acceleration features, there are algorithms for step detection under different motion patterns of pedestrians [42] and unconstrained conditions of devices [43]. For step-length estimation, linear step length (LSL) model [44] and nonlinear step length (NSL) model [45], [46] are mainly included. In addition, deep learning has been used for step estimation [47], [48] to overcome the limitations that require parameters such as the height of the user. Attitude determination was developed earlier in the field of flight control and unmanned vehicles and usually uses quaternions to solve the rotation angle in 3-D space [49]. However, there are still many problems with smart device-based attitude determination due to the limited accuracy of consumer-grade sensors [50].

Considering the relationship between cost and accuracy, our work chooses to use the peak detection method [40] as step detection algorithm, NSL model [45] as step-length algorithm, and use EKF fusion inertial sensor and magnetometer to obtain an accurate heading angle. (Detailed explanation is shown in Section III.)

1) *RNN-Based Algorithms*: RNN is a class of ANNs that are helpful in processing sequence data or time series data. Jang et al. [49] used geomagnetic data to train RNN for localization. Their RNN model uses 200 neurons, an adaptive moment estimation (Adam) optimizer, and mean square error (MSE) as the loss function. In their work, the average positioning error is 1.062 m. However, the original RNN is subject to gradient vanishing and gradient explosion. Thus, it is unable to grasp the nonlinear relationships over a long time [50]. The improved RNN model LSTM is widely used to design indoor navigation algorithms. LSTM uses gates to control the flow of information, which is needed to predict the output in the network. It can mine information from longer sequence data. Hoang et al. [31] investigated the LSTM with five structures for Wi-Fi RSS fingerprinting positioning, the multiple previous predicted locations for the input and multiple locations for the output (P-MIMO) structure LSTM with ten time steps and two hidden layers achieves the best performance, and the average localization error is 0.75 m with 80% of the errors under 1 m. This performance is better than the feedforward neural network and other conventional methods.

In this article, we use LSTM to reduce the PDR cumulative localization error by considering more historical data. (Detailed theoretical analysis is shown in Section IV-D1.)

III. METHODOLOGY

This article focuses on the positioning of pedestrians walking in a 2-D plane. Section III-A focuses on the EKF-based attitude-determination algorithm, Section III-B provides details about Wi-Fi fingerprinting positioning, while Section III-C explains the improved fusion framework using neural networks.

A. Inertial Navigation System (INS) and Magnetometer-Based Attitude-Determination EKF

The traditional PDR algorithm obtains the relative rotation angle of the device by integrating the angular velocity of the gyroscope, but the integration will lead to the gyroscope drift error accumulation. Thus, the PDR algorithm can only

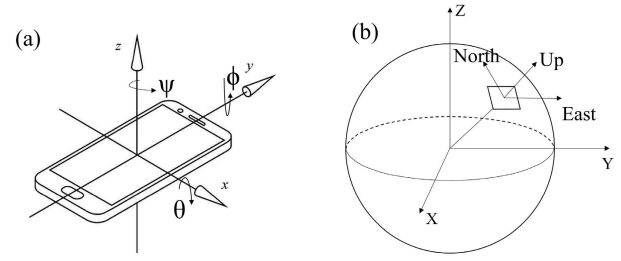


Fig. 1. Phone and navigation coordinate system. (a) Phone coordinate system. (b) ENU navigation system.

provide stable results in the short term. The corresponding magnetometer can detect the angle between the current device and the geomagnetic north pole by detecting the amount of ambient magnetic field, which can provide a long-term stable heading angle, but the complex real electromagnetic environment makes the angle fluctuate greatly in a short period.

This article refers to [51] and uses EKF to fuse the heading angles solved by the two types of sensors to provide stable and accurate attitude angles.

The coordinate of the smartphone is defined as the carrier coordinate system, which is denoted by “*b*.” The local Cartesian coordinate system east-north-up (ENU) is defined as the navigation coordinate system, which is denoted by “*n*.” The attitude of the smartphone during its motion includes the pitch angle θ , roll angle ϕ , and heading angle ψ . The navigation and phone coordinate systems are shown in Fig. 1.

The initial Euler angles (θ_0 , ϕ_0 , and ψ_0) are first calculated from the accelerometer and the magnetometer as follows (detailed derivation is shown in the Appendix):

$$\begin{aligned}\theta_0 &= \arcsin\left(\frac{a_y}{\sqrt{a_x^2 + a_y^2 + a_z^2}}\right) \\ \phi_0 &= \arctan\left(\frac{-a_x}{a_z}\right) \\ \begin{bmatrix} m_x^d \\ m_y^d \end{bmatrix} &= \begin{bmatrix} m_x^b c(\phi_0) + m_z^b s(\phi_0) \\ m_x^b s(\phi_0) s(\theta_0) + m_y^b c(\theta_0) - m_z^b c(\phi_0) s(\theta_0) \end{bmatrix} \\ \psi_0 &= \arctan\left(\frac{m_y^d}{m_x^d}\right)\end{aligned}\quad (1)$$

where a_x , a_y , and a_z are the triaxial data of the accelerometer, and s and c denote the sin and cos functions, respectively. m_{axis}^{coord} is the axis value of the magnetometer in the corresponding coordinate system. To facilitate the operation, the Euler angles are converted into quaternions (q_1 , q_2 , q_3 , and q_4), and attitude-determination EKF is also based on quaternions for data fusion.

First, the process noise covariance matrix and the observation noise covariance matrix are initialized. Assuming that the noise between each axis is correlated and independent, so they are written in the following form:

$$\begin{aligned}Q &= \begin{bmatrix} \omega_n & 0 \\ 0 & \omega_{bn} \end{bmatrix} \\ R &= \begin{bmatrix} \omega_a & 0 \\ 0 & \omega_m \end{bmatrix}\end{aligned}\quad (2)$$

where ω_n is the gyroscope noise, ω_{bn} is the gyroscope bias noise, ω_a is the accelerometer measurement noise, and ω_m is the electronic compass measurement noise.

The attitude-determination EKF consists of two phases: prediction and update. In the prediction phase, the system model is obtained by discretizing the gyroscope quaternion differential equation

$$\begin{bmatrix} q_{k+1} \\ \omega_{b(k+1)} \end{bmatrix} = \begin{bmatrix} I + \frac{1}{2}\Omega_{nb}^n T & 0 \\ 0 & I \end{bmatrix} \begin{bmatrix} q_k \\ \omega_{bk} \end{bmatrix} \quad (3)$$

where k is the k th state, Ω_{nb}^n represents the projection of the angular velocity of the carrier coordinate system to the navigation coordinate system, derived from the quaternion's properties [52], [53], I is the unitary matrix, and T is the sampling interval. Assuming

$$\begin{bmatrix} q_k \\ \omega_{bk} \end{bmatrix} = x_k, \quad \begin{bmatrix} q_{k+1} \\ \omega_{b(k+1)} \end{bmatrix} = g(x_k)$$

the Jacobian matrix of the system model is $G_k = ((\partial g(x_k))/\partial x_k)$, thus the covariance prediction equation P_k^- is

$$P_k^- = G_k P_{k-1} G_k^T + Q_{k-1}. \quad (4)$$

The measurement model is built through the use of quaternion results for magnetometer and accelerometer

$$h_k(\hat{x}_k^-) = \begin{bmatrix} 2q_1q_3 - 2q_0q_2 \\ 2q_2q_3 + 2q_0q_1 \\ q_0^2 - q_1^2 - q_2^2 + q_3^2 \\ \hat{m}_y^n(2q_1q_2 + 2q_0q_3) + \hat{m}_z^n(2q_1q_3 - 2q_0q_2) \\ \hat{m}_y^n(q_0^2 - q_1^2 + q_2^2 - q_3^2) + \hat{m}_z^n(2q_2q_3 + 2q_0q_1) \\ \hat{m}_y^n(2q_2q_3 - 2q_0q_1) + \hat{m}_z^n(q_0^2 - q_1^2 - q_2^2 + q_3^2) \end{bmatrix} + \begin{bmatrix} \omega_{a3 \times 1} \\ \omega_{m3 \times 1} \end{bmatrix} \quad (5)$$

where $\hat{m}_y^n$ and $\hat{m}_z^n$ are the projections of the normalized magnetic field data in the navigation coordinate system. Taking partial derivatives of the measurement model, the Jacobian matrix H_k can be obtained. Thus, the calculation of Kalman gain, the update of posterior state, and the posterior covariance matrix are

$$K_k = \frac{P_k^- H_k^T}{H_k P_k^- H_k^T + R} \quad (6)$$

$$\hat{x}_k = \hat{x}_k^- + K_k(z_k - h_k(\hat{x}_k^-)) \quad (7)$$

$$P_k = (I - K_k H_k) P_k^- \quad (8)$$

where z_k is the normalized quaternion of acceleration and magnetometer data in the navigation coordinate system.

By converting the EKF filtered quaternions into Euler angles, the precise and stable heading of the pedestrian can be obtained.

B. Wi-Fi Fingerprinting Positioning

In this article, we choose the K -PLDA algorithm as the Wi-Fi fingerprint localization algorithm, which is one of our previous works [39], and the algorithm can maintain high accuracy in complex pedestrian environments. This algorithm consists of an offline training phase and an online location determination phase, which are introduced separately.

1) *Offline Training Phase*: In this phase, it is necessary to establish a coordinate system in the positioning area, set up RPs, and mark them according to the appropriate interval distance. A surveyor is asked to hold the smartphone horizontally and collected RSS at each RP in different flow states. Considering each RP as a class and the RSS of different flow states under the same RP as different samples of the class, then the fingerprint database has the following form:

$$\begin{cases} D_i^c = [\text{RSS}_{ij}, (\alpha_i, \beta_i), c] \\ \text{RSS}_{ij} = \begin{bmatrix} \text{RSS}_{i1} & \cdots & \text{RSS}_{id1} \\ \vdots & \ddots & \vdots \\ \text{RSS}_{ij} & \cdots & \text{RSS}_{dj} \end{bmatrix} \\ c \in (1, 2, \dots, n) \end{cases} \quad (9)$$

where D_i^c denotes the i th sample of class c , (α_i, β_i) is the i th location of RP, and RSS_{ij} is the RSS of the i th AP in the j th flow state. The fingerprint database can also be used to train the K -PLDA model parameters.

2) *Online Location Determination Phase*: During the online location determination phase, the locations of test points (TPs) are determined by calculating the posterior probabilities in the improved PLDA hidden space. The RSS vector of the p th TP can be expressed as x^p , which is projected to become u^p . The RSS vector of the c th RP is x_i^c , which is projected to become u_i^c ; then, the posterior probability can be calculated as follows:

$$P(u^p | u_{1,\dots,n}^c) = N\left(\frac{n\Psi}{n\Psi + I} \bar{u}^c, I + \frac{\Psi}{n\Psi + I}\right) \quad (10)$$

where $\bar{u}^c = (1/n)(u_1^c + \dots + u_n^c)$, and Ψ is the orthogonal eigenvectors matrix of between-class covariance matrices. Normalizing the result of $P(u^p | u_{1,\dots,n}^c)$, the probabilities that the TP belongs to each RPs are $P = (p_1, p_2, \dots, p_i)$. The first K maximum probabilities are chosen to predict the coordinate of the TP $(\chi_k^{\text{Wi-Fi}}, \gamma_k^{\text{Wi-Fi}})$, which can be calculated as

$$(\chi_k^{\text{Wi-Fi}}, \gamma_k^{\text{Wi-Fi}}) = \sum_{i=1}^K P_i \times (\alpha_i, \beta_i). \quad (11)$$

C. Improved Fusion Framework

The previous PDR and Wi-Fi-based EKF fusion method uses the PDR error model derived from (12) as the system model and the Wi-Fi positioning results as the observation model. PDR is the relative means of determining a new position $(\chi_k^{\text{PDR}}, \gamma_k^{\text{PDR}})$ from the previous position $(\chi_{k-1}^{\text{PDR}}, \gamma_{k-1}^{\text{PDR}})$ and the state of motion (e.g., heading angle, step length) by

$$\begin{bmatrix} \chi_k^{\text{PDR}} \\ \gamma_k^{\text{PDR}} \end{bmatrix} = \begin{bmatrix} \chi_{k-1}^{\text{PDR}} + L_k \cdot \cos(\theta_k) \\ \gamma_{k-1}^{\text{PDR}} + L_k \cdot \sin(\theta_k) \end{bmatrix} \quad (12)$$

where L_k and θ_k are the step length and heading, and the subscript k and $k-1$ indicate the count of steps. Davidson and Takala [40] and Scarlett [45] give the details about step detection and step-length estimation, respectively. A nonlinear step length model that considers both maximum and minimum acceleration is adopted

$$L = \kappa \times \sqrt[4]{a_{\max} - a_{\min}} \quad (13)$$

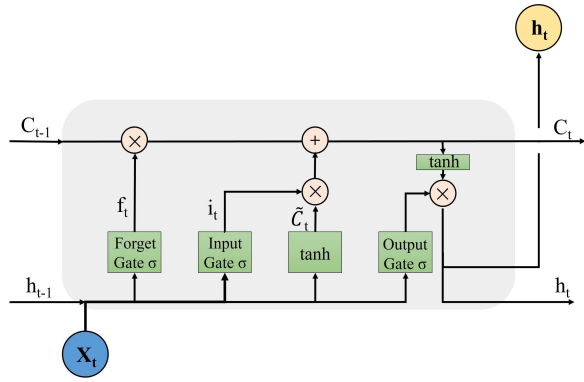


Fig. 2. LSTM block mechanism diagram. X is the input data, h is the output, f represents the information that the cell state needs to forget, while i represents the information that the cell state needs to remember, $\tilde{C}$ is a vector of new candidate values which could be added to the cell state, both σ and $\tanh$ are the activation functions, and t and $t - 1$ indicate the count of states.

where κ is a parameter related to the height of the pedestrian, which can be prelearned during a precalibration stage. $a_{\max}$ and $a_{\min}$ are the maximum and minimum acceleration, respectively.

The prediction of the PDR error model for the current location only considers one previous state, which can result in errors accumulation and the waste of historical data. In the case of indoor localization, the pedestrian can only move along a continuous trajectory, and the motion state (i.e., position, step length, and heading angle) at the current step is often similar to the previous steps. (Detailed theoretical analysis is shown in Section IV-D1.) Therefore, LSTM exploits the historical motion state data to enhance the accuracy of the localization.

LSTM is a kind of neural network containing blocks, as shown in Fig. 2. The memory and forgetting gates (σ) in the blocks are able to update the cell state (C) and select the information to be remembered and forgotten, while the output gates decide which information can be output.

The development of our PDR LSTM method is divided into two phases: a training phase and a testing phase. In the training phase, when the step detection algorithm detects a stride and the features solved by the PDR during the stride (i.e., step-length, sine, and cosine values of the heading angle and 2-D coordinates) are defined as motion state vectors

$$M = \{\chi, \gamma, L, \sin(\theta), \cos(\theta)\}. \quad (14)$$

The length of the time step severely affects the performance of the LSTM, and a large time step will incorporate historical information with little correlation and thus cause the accumulation of positioning errors. The optimal time step value will be chosen by the experiment in Section IV.

Fig. 3 shows the proposed LSTM structure. This structure has multiple inputs single output structure, where several motion state vectors from the previous steps are fed into the network. In the last step, the LSTM output layer is fully connected through the BP network to output the predicted position.

The aim of LSTM training is to minimize the loss function $L(l, \tilde{l})$ defined as the Euclidean distance between the target l

and the output $\tilde{l}$ using the BP algorithm [54]

$$L(l, \tilde{l}) = \|l - \tilde{l}\|_2. \quad (15)$$

In the testing phase, the motion vectors from the test set are fed into the pretrained LSTM to output the localization results. In order to use EKF for fusion, the state model for LSTM localization needs to be obtained. However, the neural network is a black-box system and the expression of the state model display cannot be known. Here, we consider that the LSTM location results are modified based on PDR localization, so the following equation can be approximated as the system model for EKF:

$$\begin{bmatrix} \chi_k^{\text{LSTM}} \\ \gamma_k^{\text{LSTM}} \end{bmatrix} = \begin{bmatrix} \chi_{k-1}^{\text{PDR}} + L_k \cdot \cos(\theta_k) + \varepsilon_{\chi k} \\ \gamma_{k-1}^{\text{PDR}} + L_k \cdot \sin(\theta_k) + \varepsilon_{\gamma k} \end{bmatrix} \quad (16)$$

where $\varepsilon_{\chi k}$ and $\varepsilon_{\gamma k}$ are the constant term biases that can be obtained from $\chi_k^{\text{LSTM}} - \chi_k^{\text{PDR}}$ and $\gamma_k^{\text{LSTM}} - \gamma_k^{\text{PDR}}$, respectively.

The measurement model of EKF is built through the use of Wi-Fi results

$$\begin{bmatrix} \chi_k^{\text{observe}} \\ \gamma_k^{\text{observe}} \end{bmatrix} = \begin{bmatrix} \chi_k^{\text{Wi-Fi}} + \eta \\ \gamma_k^{\text{Wi-Fi}} + \eta \end{bmatrix} \quad (17)$$

where η is the measurement noise, and $(\chi_k^{\text{observe}}, \gamma_k^{\text{observe}})$ is the observed coordinate of the k th state. After initializing the system noise σ_{PDR} and the measurement noise, the EKF can be run for fusion localization. However, since (16) is not an accurate system model, running the EKF will produce large computational errors and correspondingly large errors in state estimation, further resulting in loss of the ability to regulate the system and eventually leading to divergent results.

To avoid this situation, the BP neural network is employed to design the fusion framework. In the training phase, the LSTM positioning results, Wi-Fi positioning results, and Kalman gains are used as input features to train the two BP neural networks (horizontal and vertical coordinates fusion models, respectively). The outputs are the fused position $(\chi^{\text{fused}}, \gamma^{\text{fused}})$. We define this coordinate fusion method as BP-Kalman. Relying on the mapping approximation and self-learning capabilities of BP neural networks, more accurate localization results can be obtained. We define this fusion method as NN-EKF, and the structure of this method is shown in Fig. 4.

The NN-EKF requires multiple neural networks, which can significantly increase the training cost and the risk of over-fitting. Therefore, we design another method to fuse LSTM and Wi-Fi localization at the algorithm level. Wi-Fi predicted locations often appear within the location area, but they always suffer from RSS fluctuations in indoor environments. LSTM can reduce the cumulative error of PDR, but its predicted locations may appear in areas that are inaccessible to pedestrians (e.g., inside the walls, outside the building). In addition, the low sampling rate and violent signal fluctuations during the movement will make Wi-Fi positioning match distant RPs and thus affecting the positioning accuracy. Therefore, we further use the LSTM results to determine the Wi-Fi RPs search range, the first R RPs that are closest to the LSTM localization results will be used for Wi-Fi matching to provide more reasonable locations, and we named this fusion method as limited Wi-Fi

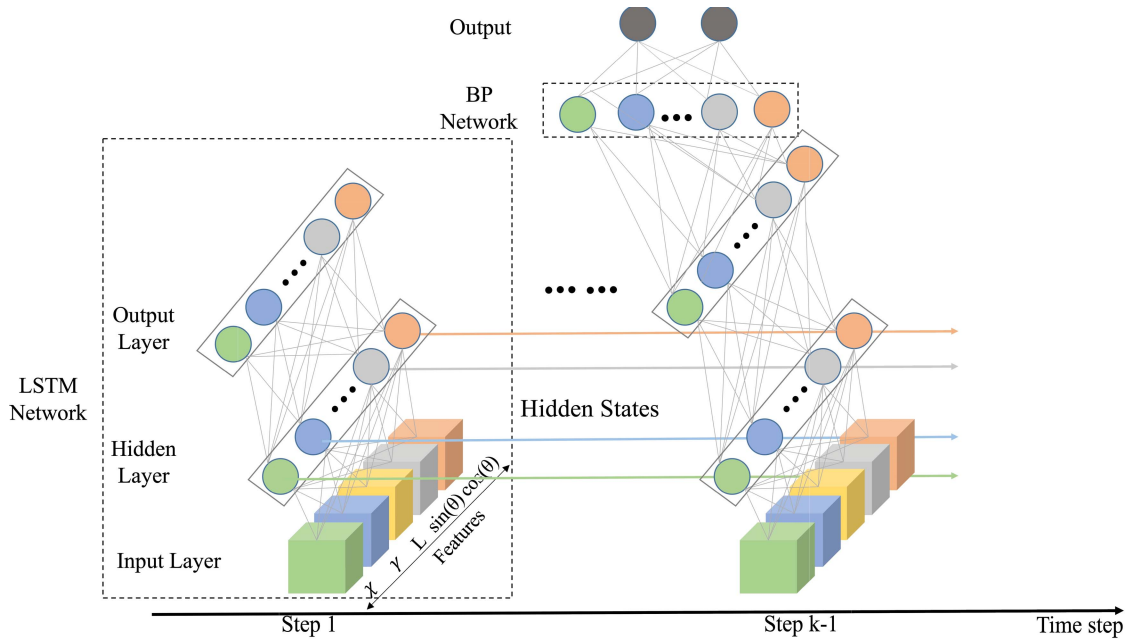


Fig. 3. Schematic of the proposed LSTM structure.

TABLE I
SPECIFICATIONS OF INERTIAL SENSORS IN TESTED SMARTPHONES

	ICM-40607(InvenSense)	ICM-20690(InvenSense)
Gyro.	At 25°C ± 1%	At 25°C ± 1%
	Thermal drifts: ±0.01 %/ °C (-40 to + 85 °C)	Thermal drifts: ±0.01 %/ °C (-40 to + 85 °C)
Biases	ZRO Variation: ±0.03 %/ °C (-40 to + 85 °C)	ZRO Variation: ±0.01 %/ °C (-40 to + 85 °C)
Accel.	At 25°C ± 1%	At 25°C ± 1%
	Thermal drifts: ±0.01 %/ °C (-40 to + 85 °C)	Thermal drifts: ±0.008 %/ °C (-40 to + 85 °C)
Biases	ZRO Variation: ±0.5 mg/ °C (-40 to + 85 °C)	ZRO Variation: ±0.5 mg/ °C (-40 to + 85 °C)

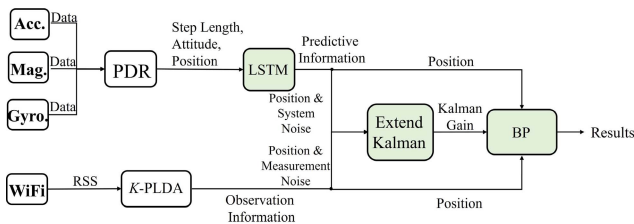


Fig. 4. Structure of NN-EKF.

method. The structure of the two fusion methods is shown in Fig. 5.

IV. EXPERIMENT AND ANALYSIS

A. Experimental Platforms and Setup

The experiment platform can be any Android device with accelerometer, gyroscope, magnetometer, and Wi-Fi receiver. All sensor data were collected through an app developed by Android Studio, and the sampling frequency of the inertial sensor was set to 66 Hz, while the Wi-Fi sample rate was 0.3 Hz. Due to the restrictions of the hardware and Android system, the Wi-Fi update frequency can only be set to a

maximum of 0.3 Hz. For the problem of the inconsistent sampling frequency, we made replace the missing value of the Wi-Fi signal by copying the value of the last update.

Two different smartphones were used, Table I shows the parameters of the inertial sensors they are equipped with, in practice, low-cost MEMS inertial sensors may suffer from severe operational bias and thermal drift, and we performed a simple manual calibration of the two phones refer to [55].

B. Test Area and Database Generation

The sixth floor on the south wing of the School of Physics and Electronics at Central South University (CSU) was selected as test area, as shown in Fig. 6, the size of this area is around $50 \times 16 \text{ m}^2$, and there is a main corridor which is 40 m long and a lobby which is $8 \times 9 \text{ m}^2$. The test area is covered by an 802.11 WLAN with up to 13 stable APs. We set up 85 RPs and each RP is spaced 1.2 m apart. The RSS was collected at three different flow states with a sampling interval of 2 s, and each AP collected 60 samples and took their mean values to build the fingerprint database. For some AP signals that are too weak to be detected, a minimum fixed value is set to -100 dBm [56].

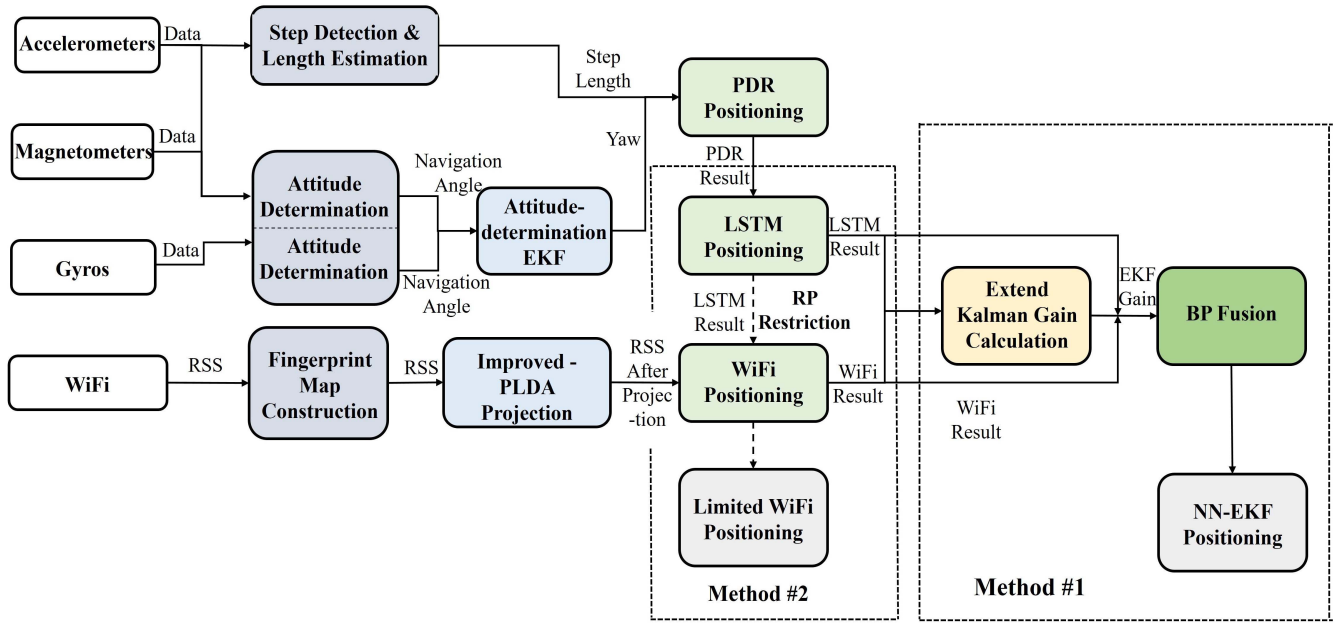


Fig. 5. Structure for PDR and Wi-Fi integration.

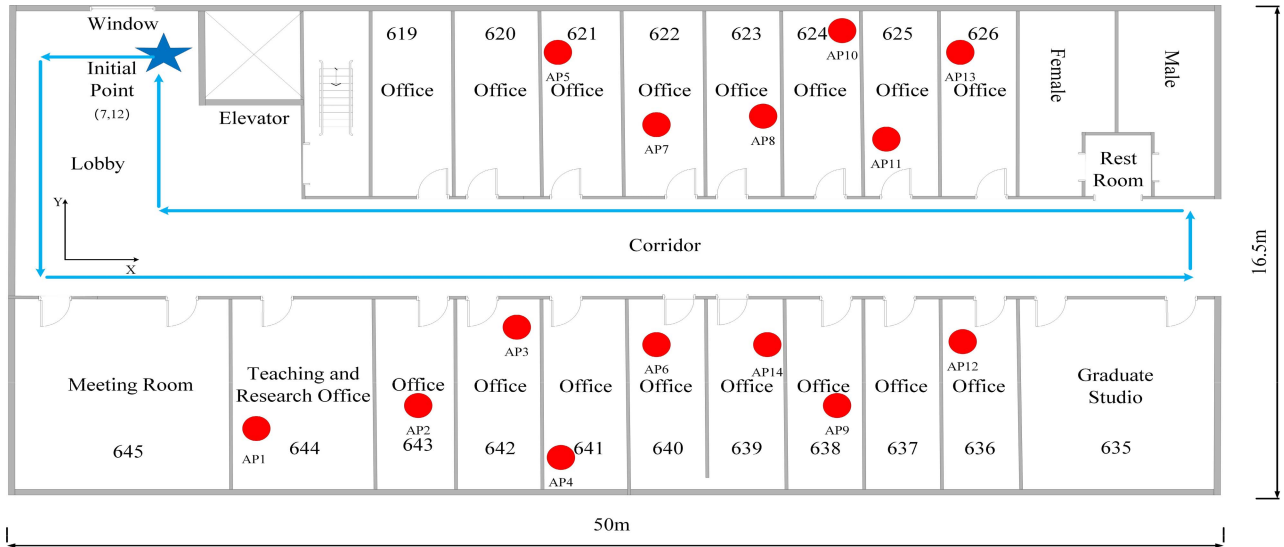


Fig. 6. Floor map of the test site. The blue solid line is the mobile user's walking trajectory with arrows pointing toward the walking direction.

The database used for training the LSTM consists of 100 random trajectories (2002 steps in total), where the length of the trajectory is between 10 and 100 m. This database was also used for Wi-Fi localization, and then, the LSTM localization results and Wi-Fi localization results were used to train the BP neural network.

C. Test Data Collection

A total of two testers performed the data collection. During the experiment, testers each held a smartphone in front of their chest, so that the phone's y-axis pointed directly ahead and the z-axis pointed opposite to gravity. The testers were asked to walk sequentially on a predesigned trajectory at a normal speed. The preset trajectory is 112 m long, including five turns and six straights, which are consistent with the daily activity trajectory of most staff. The preset trajectory is shown as the blue solid line in Fig. 6.

Since the initial position of the PDR algorithm needs Wi-Fi positioning to determine and the accuracy of the initial position will greatly affect the subsequent positioning process of PDR, we require the experimenter to stay at the initial point for at least 1 min to obtain abundant and stable RSS to improve the positioning accuracy of the initial point.

D. Hyperparameter Setting and Analysis

This section provides a detailed study of selecting the optimal hyperparameter for the proposed fusion methods and the setting of EKF parameters. Results of parameter selection for neural networks in the following are obtained after fivefold tests with random training trajectories.

First, we select the hyperparameter for LSTM and BP neural network.

1) *Different Time Steps*: Fig. 7 compares the cumulative probability distribution (CDF) of the localization error of

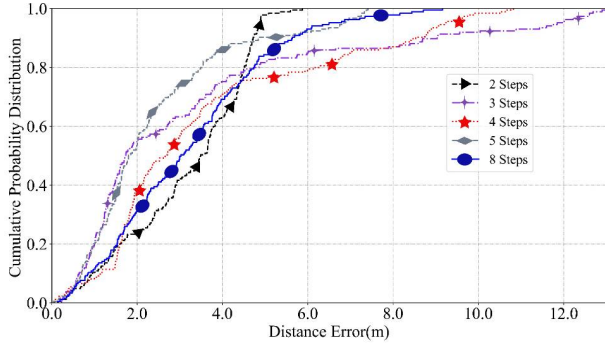


Fig. 7. CDF of the localization error of different steps in LSTM structure.

different time steps in LSTM structure, i.e., different training trajectories containing 2, 3, 4, 5, or 8 time steps, respectively. Two steps and five steps trajectories consistently show a better accuracy with 90% of the errors within 4.80 m. Furthermore, the maximum error of two steps training trajectory is 5.88 m, which is lower than 7.41 m obtained from five steps. In the rest of this article, two steps are chosen to train LSTM.

The theoretical analysis is as follows. When the movement patterns of pedestrians are relatively similar within a certain number of steps (e.g., straight ahead), the movement vectors do not change much between the front and back parts, and then, the time series has a certain regularity, while if the pedestrians' movement patterns change a lot (e.g., turning), it is difficult to find the regularity and thus difficult to use LSTM for accurate prediction. This article uses cosine similarity [57] to measure the similarity of pedestrian movement patterns. Compared with the distance-based measures, the cosine similarity is not sensitive to the value but more distinguishes the difference from the direction, and the greater the cosine similarity, the higher the similarity of the two trends, so it can be a good measure of the similarity of pedestrian movement patterns. The calculation of the cosine similarity of two motion vectors can be given as follows:

$$\cos = \frac{A \cdot B}{\|A\| \cdot \|B\|} = \frac{\sum_{i=1}^n A_i \times B_i}{\sqrt{\sum_{i=1}^n (A_i)^2} \times \sqrt{\sum_{i=1}^n (B_i)^2}} \quad (18)$$

where A and B are the motion vectors, and A_i and B_i are the i th components of A and B , respectively.

We selected eight training trajectories (four straight trajectories and four trajectories containing turns) from the database to investigate how similar the pedestrian's motion vector at moment t is to the previous eight-step motion vector. To facilitate the demonstration of variability, we subtracted the cosine similarity between the vectors by 1 to thereby obtain the cosine variance matrix. As shown in Fig. 8, for the straight trajectories, the difference between the motion vectors of the first eight steps and the current step is not large, which indicates that the motion pattern of the pedestrian does not change basically when walking straight, so using longer historical data can make the prediction more accurate in this motion pattern. For the trajectory that includes turning, the difference between the current step and the first two steps is small, but the difference between the motion vector and the two steps before is large, so choosing a shorter history data

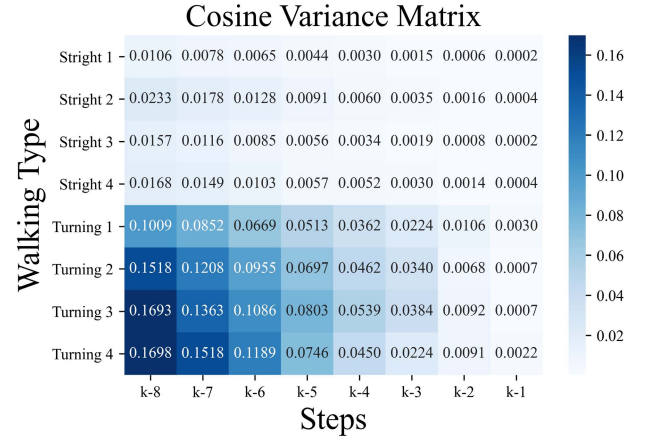


Fig. 8. Cosine variance matrix.

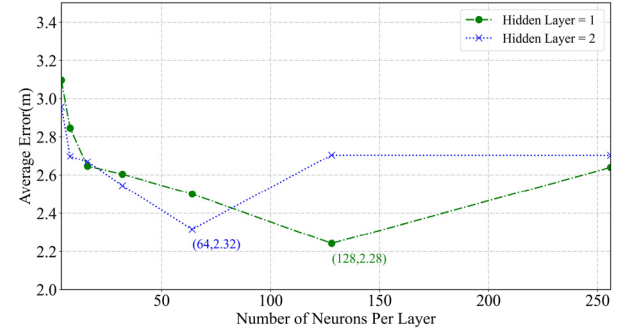


Fig. 9. Average localization errors of LSTM with the different numbers of hidden layers and neurons per layer.

can make the prediction more accurate, which explains to some extent that using the historical motion trajectory of two and five steps can make the prediction accuracy relatively high, but when the pedestrian motion pattern includes turning, the prediction using the first five steps data will have a large error, which also explains that the maximum error of prediction using two steps is smaller than that of prediction using five steps.

2) *Different Layers and Neurons*: Fig. 9 illustrates the average localization errors of LSTM with the different numbers of hidden layers and neurons per layer. In general, adopting one hidden layer can achieve satisfactory positioning accuracy. Two hidden layers and 64 neurons per layer result in the best accuracy of 2.32 ± 0.84 m. In comparison, the best accuracy of one hidden layer is 2.28 ± 0.51 m when the number of neurons per layer is 128. Increasing more neurons will lead to overfitting and reduce the localization accuracy. The best average error of two hidden layers is about the same as that of one hidden layer, but using two hidden layers will greatly increase the cost of training the model. To reduce the risk of overfitting, 1 hidden layer and 128 neurons per layer are the optimal parameters for our proposed method.

3) *Learning Rate, Optimization Algorithm, and Dropout Rates for LSTM*: Table II shows the effect of different optimization algorithms and learning rates on the average localization error. In general, Adam optimization is better than stochastic gradient descent (SGD) algorithm. Adam with the learning rate of 0.01 provides the minimal positioning errors of 2.23 ± 0.42 m. A smaller learning rate will lead to overfitting,

TABLE II
DIFFERENT LEARNING RATES AND OPTIMIZATION ALGORITHMS

Optimization algorithm	Learning rate	Average error(m)
Adam	0.001	3.18±0.46
Adam	0.01	2.32±0.42
Adam	0.1	2.90±0.76
SGD	0.001	3.75±1.21
SGD	0.01	4.98±1.14
SGD	0.1	8.84±1.19

TABLE III
DIFFERENT DROPOUT RATES

Dropout rate	Average error(m)
0.1	3.38±0.66
0.2	3.33±0.68
0.3	3.07±0.74
0.4	2.88±0.54
0.5	2.26±0.56
0.6	3.57±0.72

while a larger learning rate will lead to underfitting, both of which will increase the average localization error. Table III demonstrates average positioning error at different dropout rates, with the dropout rate equal to or less than 0.2, there is no significant difference in positioning results, i.e., 3.38 ± 0.68 m at a dropout rate of 0.1 and 3.33 ± 0.64 m at a dropout rate of 0.2, when the dropout rate increases to 0.5, and the localization error reaches the bottom of 2.26 ± 0.56 and then rises again as the dropout rate increases.

When training the BP network, since the input features contain the prediction of the same location by LSTM and Wi-Fi, and the real location corresponding to the training set has been used by LSTM, the resulting collinearity between predictors will cause overfitting. In order to reduce the risk of overfitting, this research looks for suitable parameters from a simple neural network structure.

4) *Different Neurons Per Layer for BP*: Fig. 10 shows that for one hidden layer, the fusion result is better than the single localization result (LSTM is 2.26 m and Wi-Fi is 3.41 m) when the number of neurons per layer is within 12, especially when the number of neurons per layer is 8 or 10, a better fusion result can be obtained (1.66 and 1.67 m, respectively), and increasing the number of neurons afterward will increase the localization error. In order to reduce the risk of overfitting, one hidden layer with eight neurons is chosen as the BP network structure in this article.

5) *Learning Rate and Optimization Algorithm for BP*: Table IV compares the performance of BP networks with different optimization algorithms and learning rates. Using the SGD optimization algorithm with a small learning rate can improve the localization effect to some extent, but its accuracy is not as high as Adam with a learning rate of 0.1. Therefore,

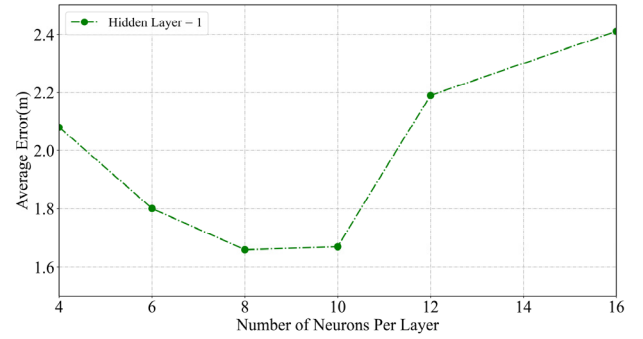


Fig. 10. Average localization errors of BP-Kalman with the different neurons per layer.

TABLE IV
DIFFERENT LEARNING RATES AND OPTIMIZATION ALGORITHMS

Optimization algorithm	Learning rate	Average error(m)
Adam	0.001	2.25±0.22
Adam	0.01	1.67±0.18
Adam	0.1	1.36±0.26
SGD	0.001	1.92±0.31
SGD	0.01	4.41±0.19
SGD	0.1	4.39±0.24

TABLE V
INITIAL SETUP PARAMETERS FOR LSTM AND BP NETWORK

Category	Value	Value
NN type	LSTM	BP
Time step	2	/
Loss function	RMSE	RMSE
Hidden layer (HL)	1	1
Number of neurons for each HL	128	8
Dropout	0.5	/
Optimizer	Adam	Adam
Learning rate	0.01	0.1

Adam with a learning rate of 0.1 is the optimal parameter for our proposed fusion method. The initial parameter settings of LSTM and BP neural network are shown in Table V.

6) *Setting of the EKF Parameters*: Indoor environments are complex and unpredictable. Thus, although various devices are used, the same set of EKF parameters was set to evaluate the robustness of the algorithm. The EKF parameters include the initial state vector $[\omega_n, \omega_{bn}, \omega_a, \omega_m]$ for the attitude-determination EKF and the system noises (σ_{PDR}) and the measurement noises (n) for the NN-EKF. The values of the parameter initialization are shown in Table VI, where the parameters of the attitude-determination EKF are obtained by actually recording the fluctuations of the static values of the sensors, and the parameters of the NN-EKF are obtained according to the localization of the training set.

7) *Setting of the Limited Wi-Fi Parameters*: After the K-PLDA algorithm is improved to the limited Wi-Fi method, the model parameters change from matching with the first

TABLE VI
SETTING OF THE EKF PARAMETERS

Parameters		Values
Attitude-determination EKF	ω_n	$[1e-6, 1e-6, 1e-6]$
	ω_{bn}	$[1e-8, 1e-8, 1e-8]$
	ω_a	$[1e-1, 1e-1, 1e-1]$
	ω_m	$[1e-3, 1e-3, 1e-3]$
NN-EKF	σ_{pdr}	3.5
	η	6

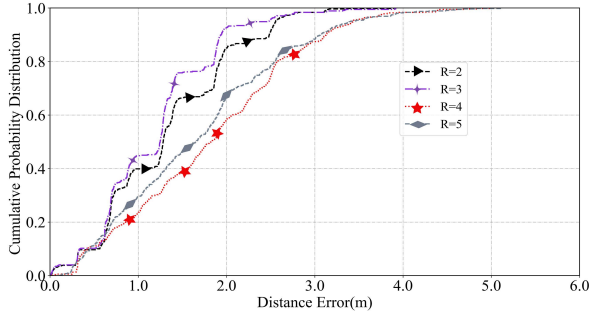


Fig. 11. CDF of the localization error of different R values in limited Wi-Fi method.

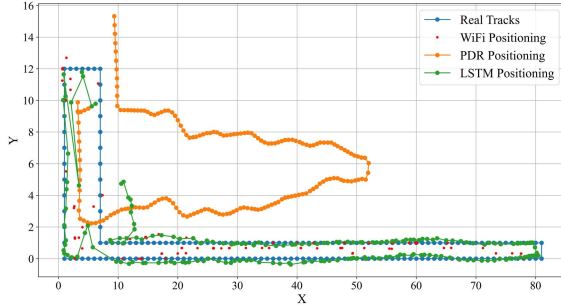


Fig. 12. Comparison of PDR trajectory, LSTM trajectory, and Wi-Fi positioning with real trajectory.

K RPs with the highest posterior probability to match with the first R RPs with the smallest Euclidean distance from the estimated position of the LSTM. The CDF diagram for positioning using different R values is shown in Fig. 11.

When $R = 3$, the highest localization accuracy can be achieved; thus, we use 3 as the number of RP matches for the limited Wi-Fi method.

E. Testing and Comparison of the Proposed Methods

Fig. 12 shows the comparison of PDR trajectory, LSTM, and Wi-Fi positioning with real trajectory. Due to the low sampling frequency and the influence of indoor complex electromagnetic environments, Wi-Fi localization does not perform well in a dynamic environment. PDR makes the localization trajectory far from the real trajectory due to the accumulation of errors, while LSTM uses historical information and powerful non-linear mapping ability to effectively reduce the accumulated errors, making the localization trajectory closer to the real trajectory.

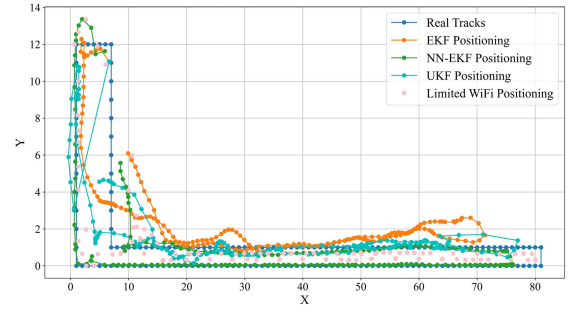


Fig. 13. Localization trajectories of different fusion methods.

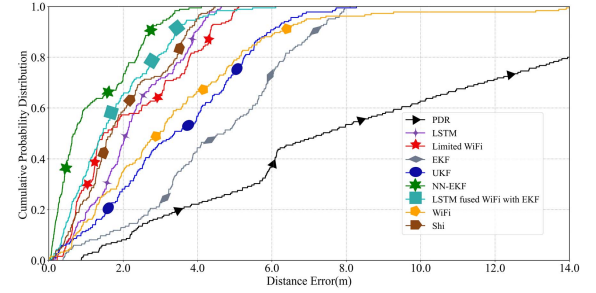


Fig. 14. CDF of position errors when using different location methods.

Fig. 13 compares localization trajectories of different fusion methods. In general, all the fusion methods make the localization trajectory closer to the real trajectory. The NN-EKF trajectory is smoother in the straight part and has less fluctuation at the turn part compared to the LSTM localization trajectory. For the limited Wi-Fi method, it provides more reasonable locations compared to Wi-Fi or PDR localization.

Fig. 14 shows the CDF of position errors when using different location methods. The NN-EKF algorithm (green line) outperforms the other algorithms especially LSTM fused Wi-Fi with EKF in terms of positioning accuracy, which illustrates that the use of the BP neural network to improve EKF for fusion is effective. Specifically, the probability of the positioning accuracy of NN-EKF within 2 m is 73.1%, which is 59.5% and 45.0% higher than EKF and UKF, respectively. The probability that the positioning accuracy is within 1 m is also significantly improved compared to other algorithms, reaching 60.5%. The probability of the accuracy of limited Wi-Fi within 2 m is also achieved at 57.2%, which is 21.6% higher than Wi-Fi.

Table VII compares the statistical values of errors (i.e., average errors, median errors, and max errors) and standard deviation of all tested positioning algorithms. Wi-Fi provided a better performance (average error 3.37 m) than PDR (average error 8.57 m), as PDR suffered from drifts. However, Wi-Fi does not perform as well in dynamic environments as it does in static environments, due to low sampling rates and RSS fluctuations. When aided by LSTM positioning results, the limited Wi-Fi average error decreased to 2.05 m.

Due to the significant cumulative errors of PDR and the poor performance of Wi-Fi in dynamic positioning, traditional fusion strategies are difficult to provide high positioning accuracy. However, when using LSTM to reduce the cumulative error of PDR and then performing EKF fusion can obtain better results (average error of 1.74 m). The two fusion

TABLE VII
COMPARISON OF THE ERROR OF DIFFERENT POSITIONING ALGORITHMS

Algorithms	Average error(m)	Median error(m)	Max error(m)	Standard deviation(m)
PDR [17]	8.57	7.38	18.80	5.12
WiFi [39]	3.37	3.00	14.58	2.51
EKF [21]	4.50	4.71	8.10	2.04
UKF [23]	3.46	3.46	8.27	1.94
Shi[58]	1.96	1.70	4.47	1.20
LSTM	2.24	2.11	4.64	1.21
Limited WiFi	2.05	1.47	5.11	1.50
LSTM fused WiFi with EKF	1.74	1.45	6.10	1.23
NN-EKF	1.18	0.70	4.10	1.05

methods proposed in this research, NN-EKF and limited Wi-Fi, reduce the average error by 73.8%, 65.9%, and 54.4%, 40.8% compared with the conventional fusion methods EKF and UKF. Their high average positioning accuracy, small maximum error, and small standard deviation provide stable and highly accurate position services in dynamic environments.

F. Stability and Robustness

Since the input features of the LSTM are solved by PDR and each person has his or her walking pattern, this will lead to differences in the step length and heading angle solved by PDR. To investigate the differences in step length and heading angle brought by different walking styles, an amount of Gaussian noise is added to step length and heading angle.

$$\begin{bmatrix} \theta' \\ L' \end{bmatrix} = \begin{bmatrix} \theta + \theta_n \\ L + L_n \end{bmatrix}$$

where θ_n and L_n are the random variables that follow Gaussian distribution

$$\begin{bmatrix} \theta_n \\ L_n \end{bmatrix} \sim \begin{bmatrix} N(0, \sigma_\theta^2) \\ N(0, \sigma_L^2) \end{bmatrix}$$

and Fig. 15 illustrates the localization errors of NN-EKF with both the no added noise data and erroneous data. It shows that the positioning accuracy is reduced using the data with added noise, but the probability of positioning error at 2 m (67.7%) is similar to the result of using the original data (72.8%). The localization accuracy does not change much with the increasing noise addition range, and the average positioning error with the addition of noise data is 1.6 ± 0.3 m, which is only about 0.4 m different from using the original data. The results indicate that our proposed NN-EKF method is robust to the localization error of the different walking styles.

In reality, a smartphone is usually held in the hand and it is not always oriented horizontally toward the front. Here, we increase the experiment when the y-axis of the phone is

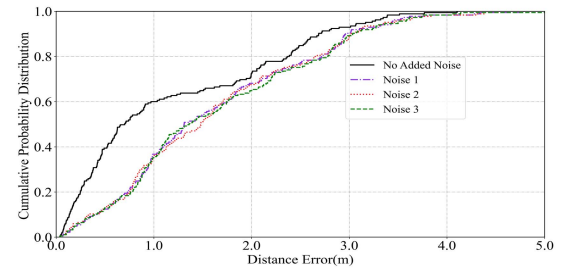


Fig. 15. CDF of NN-EKF localization errors in different Gaussian error scenarios. Noise 1 represents ($\sigma_\theta = (\pi/36)$ rad and $\sigma_L = 0.10$ m), noise 2 represents ($\sigma_\theta = (7\pi/180)$ rad and $\sigma_L = 0.15$ m), and noise 3 represents ($\sigma_\theta = (\pi/18)$ rad and $\sigma_L = 0.20$ m).

facing the top to verify the positioning performance of the algorithm.

Fig. 16 shows the CDF of NN-EKF and limited Wi-Fi for the phone in different postures. In general, the positioning accuracy is higher when the phone is held horizontally than when it is held vertically (the average localization error is 1.50 m for NN-EKF and 2.54 m for limited Wi-Fi when the phone is held vertically). Mainly because the phone is held horizontally when the training data are collected, and the RSS values will fluctuate when the phone's posture changes, thus affecting Wi-Fi positioning and fusion algorithm. However, the increase in error is still within the acceptable range (0.32 and 0.49 m for NN-EKF and limited Wi-Fi), indicating that our algorithm can provide a consistent and reliable solution under different motion conditions.

G. Calculation Complexity

To evaluate the calculation complexity of our proposed algorithms, we counted the time required when using different sizes of training data and prediction time for one sample of the test dataset. All experiments were run on a PC with Intel i5-6200U CPU and 8-GB RAM. Fig. 17 shows the training time and average prediction error corresponding to different size training data. The training time was spent mainly

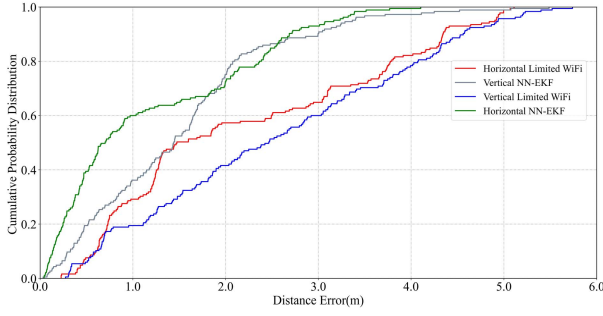


Fig. 16. CDF of NN-EKF and limited Wi-Fi for phone in different postures. Holding the phone horizontally means that the y-axis is facing straight ahead, while holding it vertically means that the y-axis is facing in the opposite direction of gravity.

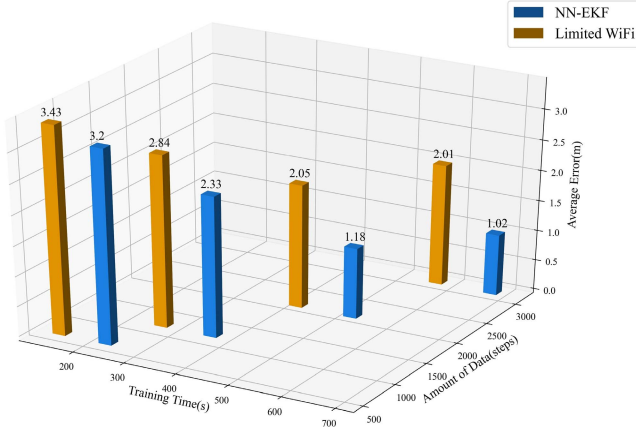


Fig. 17. Training time and average prediction error corresponding to different size training data.

TABLE VIII

CALCULATION TIME USING DIFFERENT METHODS

Method	Training Time(s)	Prediction Time for One Sample(ms)
NN-EKF	515.7	2.44
Limited WiFi	409.8	1.57

on training LSTM and BP neural networks. The increase of training data will improve the prediction accuracy, but the accuracy improvement is limited when the amount of training data increases to a certain level. We consider that this phenomenon occurs because too much data contain more repetitive trajectories of pedestrians, and it is difficult for the neural network to learn their nuances, resulting in more time spent but little performance improvement.

Table VIII lists the training time and test time of the two methods we proposed under the 2002 step training data. The NN-EKF requires a longer training time but also provides higher accuracy, and the limited Wi-Fi limits the RP matching range to speed up the online matching process, achieving fast positioning. The delay at the millisecond level for both methods does not affect the online real-time positioning.

V. CONCLUSION

This article proposed a new PDR/Wi-Fi fingerprinting integration method. The LSTM method is used to capture the

relation between a series of pedestrian historical movement states and determined user's moving path. The powerful nonlinear mapping capability of BP neural networks is able to enhance the EKF performance for the Wi-Fi data fusion. Experimental results show that the NN-EKF structure achieves the average accuracy of 1.18 m with 73% of the error less than 2 m, which outperforms EKF and UKF fusion methods. Furthermore, the main challenges of Wi-Fi method in dynamic scenes including the RSS fluctuations and the RSS short collecting time have been effectively mitigated by limited Wi-Fi fusion method, which also achieves fast positioning without excessive loss of accuracy. In addition, the analysis of LSTM, BP, and limited Wi-Fi with important parameters, such as time step, loss function, hidden layers, and calculation complexity, has been discussed in detail.

In our future work, we intend to integrate our proposed method with the algorithm for solving the phone in different motion conditions to provide a consistent reliable solution under different motion conditions.

APPENDIX

The navigation coordinate system is the ENU, and the carrier coordinate system is right, forward, and up (x, y, z), and assuming the rotation order of attitude angle is: y, x , and z (roll, pitch, and yaw).

For the accelerometer, the initial triaxial acceleration in the navigation coordinate system is $[00|g|]^T$, and the triaxial acceleration measured in the smartphone coordinate system is $[a_x a_y a_z]^T$, so we can obtain the following equation:

$$\begin{bmatrix} a_x \\ a_y \\ a_z \end{bmatrix} = C_b^n \cdot \begin{bmatrix} 0 \\ 0 \\ |g| \end{bmatrix} \quad (19)$$

where C_b^n represents the transformation matrix of the carrier coordinate system to the navigation coordinate system, and C_b^n can be given as (20), shown at the top of the next page.

Substituting (20) into (19), we can calculate θ_0 and ϕ_0

$$\theta_0 = \arcsin\left(\frac{a_y}{|g|}\right) \quad (21)$$

$$\phi_0 = \arctan\left(\frac{-a_x}{a_z}\right) \quad (22)$$

where

$$|g| = \sqrt{a_x^2 + a_y^2 + a_z^2}.$$

Assuming that the magnetometer is placed completely horizontally (we define the carrier coordinate system in this state as s) without considering the vertical direction, and considering only the XOY plane, then ψ_0 can be calculated as

$$\psi_0 = \arctan\left(\frac{m_y^d}{m_x^d}\right). \quad (23)$$

But in fact, the smartphone cannot be placed horizontally on the ground, which means there is roll and pitch. At this time, the magnetic field strength is not the true component on the X and Y axes, so it must be based on $m_x^b, m_y^b, m_z^b, \theta_0$, and

$$C_b^n = \begin{bmatrix} c(\phi_0)c(\psi_0) - s(\phi_0)s(\theta_0)s(\psi_0) & c(\phi_0)s(\psi_0) + s(\phi_0)s(\theta_0)c(\psi_0) & -c(\theta_0)s(\phi_0) \\ -c(\theta_0)s(\psi_0) & c(\theta_0)c(\psi_0) & s(\theta_0) \\ s(\phi_0)c(\psi_0) + c(\phi_0)s(\theta_0)s(\phi_0) & s(\phi_0)s(\psi_0) - c(\phi_0)c(\psi_0)s(\theta_0) & c(\phi_0)c(\theta_0) \end{bmatrix} \quad (20)$$

ϕ_0 to find the two quantities projected on the XOY plane m_y^s and m_x^s .

Since the vertical direction is not considered when the smartphone is placed horizontally, the transformation matrix C_b^d becomes

$$C_b^d = \begin{bmatrix} c(\phi_0) & 0 & s(\phi_0) \\ s(\theta_0)s(\phi_0) & c(\theta_0) & -c(\phi_0)s(\theta_0) \\ -c(\theta_0)s(\phi_0) & s(\theta_0) & c(\phi_0)c(\theta_0) \end{bmatrix}. \quad (24)$$

According to

$$\begin{bmatrix} m_x^d \\ m_y^d \\ m_z^d \end{bmatrix} = C_b^d \begin{bmatrix} m_x^b \\ m_y^b \\ m_z^b \end{bmatrix}$$

and substituting θ_0 and ϕ_0 calculated by (19) and (20) into C_b^d , we get

$$\begin{bmatrix} m_x^d \\ m_y^d \end{bmatrix} = \begin{bmatrix} m_x^b c(\phi_0) + m_z^b s(\phi_0) \\ m_x^b s(\phi_0)s(\theta_0) + m_y^b c(\theta_0) - m_z^b c(\phi_0)s(\theta_0) \end{bmatrix}. \quad (25)$$

Substituting m_x^d and m_y^d into (23), ψ_0 can be calculated.

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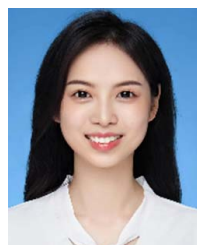
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